CO2 You are a database consultant hired by the BRACU Research Center (BRC). Their current system for tracking projects is outdated and unable to handle the complexity of their data. You are tasked with updating the existing database design given below.

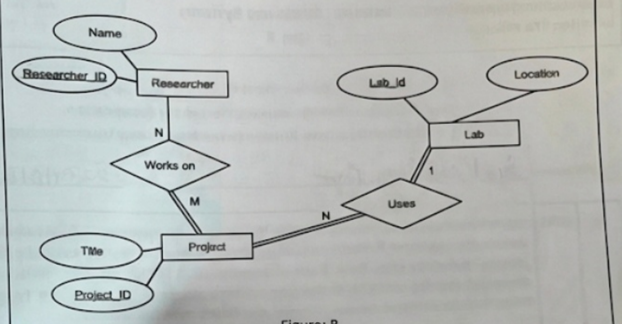
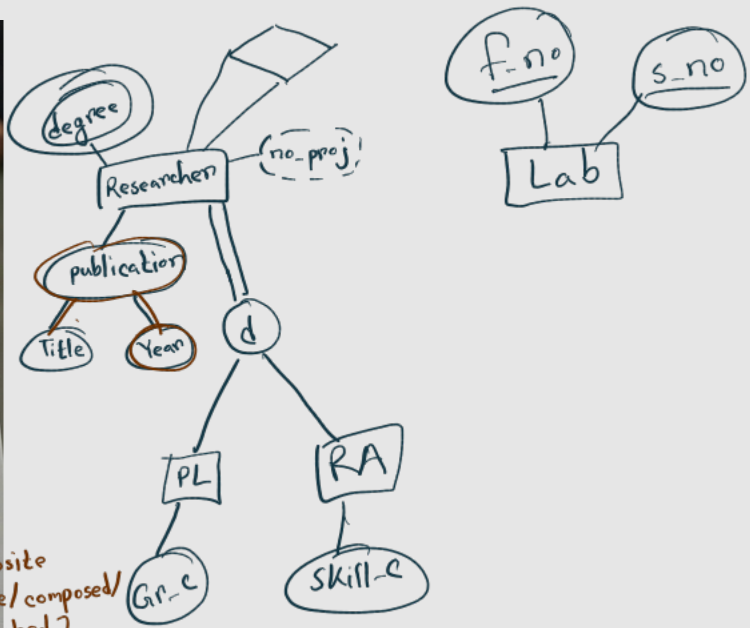
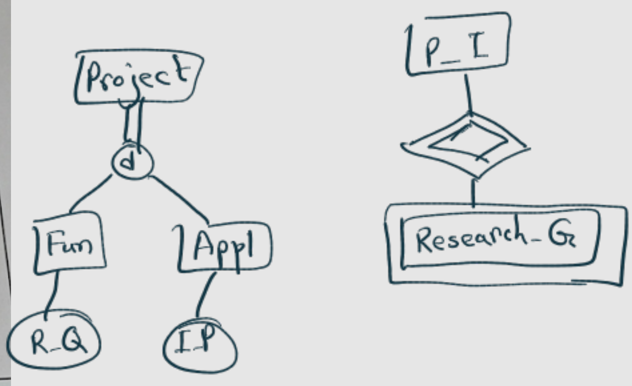


Figure: B

You have to update the above design (Figure B) according to the latest requirements given below:

- For all researchers, BRC must store their multiple academic_degrees (multivalued) and a list of all their publications, where each publication includes a Title and Year_of_Publication. (composite [include/composed/embed]) The total number_of_projects for any researcher is shown, calculated by counting the current projects they are working on. (Derived)
- Each lab is now uniquely identified by the combination of floorNumber and SerialNumber, instead of a single Lab_Id. (Key)
- Researchers are now classified as either a Principal_Investigator (PI), having a Grant_Code, or a Research_Assistant (RA), having a Skill_Certification. No other type of researchers exists other than PI or RA; All junior researchers are supervised by exactly one senior researcher. (sub class)
- The existing Project entity must be updated accordingly: the key attribute is now called Project_Code, a StartDate, and a Budget is added. Every project must be categorized as either Fundamental, having a Research Question, or Applied, having an Industry_Partner.
- All Principal_Investigators have more than one Research_Grant. A Research_Grant cannot exist on its own; it must always be associated with the specific Principal_Investigator who secured it. A Research_Grant's Grant_Number is only unique for a given PI (thus different PIs may have the same Grant_Number). Each grant also has a Funding_Amount and Funding_Source. (weak entity)





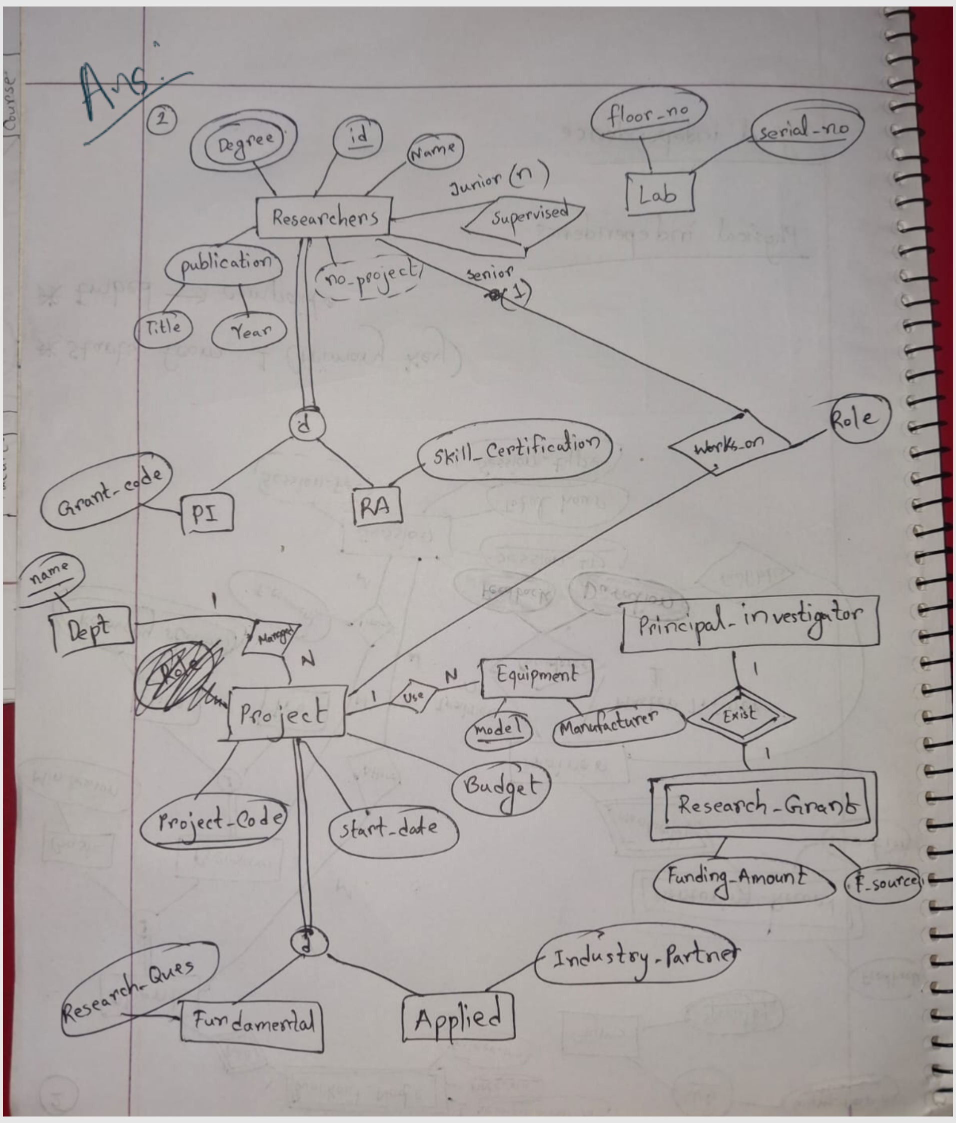
- The Works_On relationship between researchers and projects must also be modified to include a Role attribute, recording the specific role a researcher plays in each project.
- Every project must be managed by exactly one department. Each department has a unique name, and buildingNumber. Additionally, the BRC must track all Equipment, which has a unique Model_Number, and Manufacturer. A project may require multiple equipment, but the equipment can be used by only one project at a time.

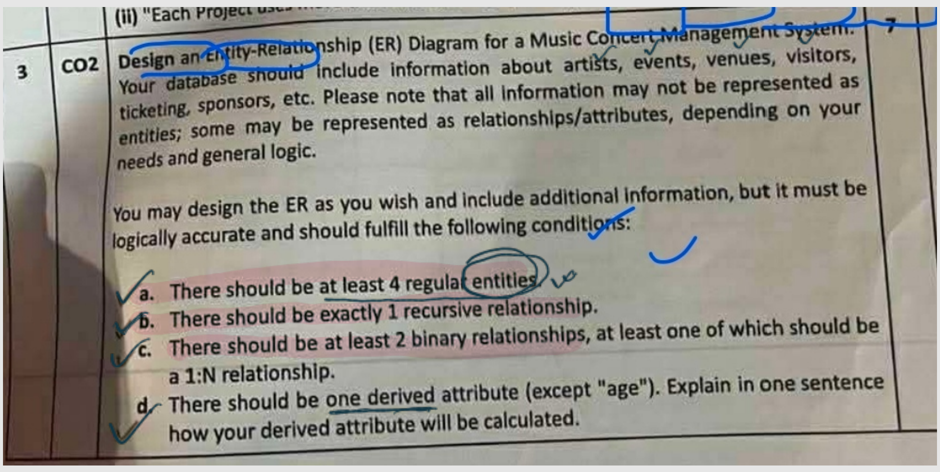
(a) Update the existing ER diagram (Figure B) to **construct** an EER diagram according to the above requirements. *Unchanged portions of the existing diagram must also be shown.* Do not assume any additional attributes, entities, relationships, keys, multivalued, or composite attributes other than those indicated in the question. 13

(b) **State and explain** the following questions with only one sentence:

(i) Does every Researcher work on a project according to the given ER diagram? [No] 1

(ii) "Each Project uses more than one Lab" - do you support or refute this statement? [True]





(ii) "Each Project us...

3	CO2	Design an Entity-Relationship (ER) Diagram for a Music Concert Management System. Your database should include information about artists, events, venues, visitors, ticketing, sponsors, etc. Please note that all information may not be represented as entities; some may be represented as relationships/attributes, depending on your needs and general logic. You may design the ER as you wish and include additional information, but it must be logically accurate and should fulfill the following conditions: a. There should be at least 4 regular entities. b. There should be exactly 1 recursive relationship. c. There should be at least 2 binary relationships, at least one of which should be a 1:N relationship. d. There should be one derived attribute (except "age"). Explain in one sentence how your derived attribute will be calculated.	7